

Smaller 5-chromatic unit-distance graphs on two icosahedral spheres

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Abstract

Voronov, Neopryatnaya and Dergachev constructed 5-chromatic unit-distance graphs on the sphere $S^2(r_1)$, $r_1 = \cos \frac{\pi}{10}$, with 372 vertices, and on $S^2(r_2)$, $r_2 = \cos \frac{3\pi}{10}$, with 972 vertices, and noted that the first graph stays 5-chromatic after deleting any single vertex. By SAT-based vertex minimization we find a 5-chromatic subgraph of the first graph with 231 vertices and 938 edges, and one of the second with 961 vertices and 4028 edges. Both are vertex-critical and contain no Moser spindle; the smaller one is the best found by our search, not a proved minimum. All coordinates are given exactly in nested radicals and verified in exact arithmetic; non-4-colourability is certified by DRAT proofs checked with DRAT-TRIM.

1 Introduction

The chromatic number $\chi(X)$ of a subset $X \subseteq \mathbb{R}^n$ is the least number of colours needed to colour X so that no two points at Euclidean distance 1 receive the same colour. For the plane, de Grey [3] proved $\chi(\mathbb{R}^2) \geq 5$; the smallest known 5-chromatic unit-distance graph in the plane has 509 vertices [7] (see also [4]).

For the sphere $S^2(r) \subset \mathbb{R}^3$ of radius $r > \frac{1}{2}$, Simmons [8] studied colourings, and Voronov, Neopryatnaya and Dergachev [10] constructed the first 5-chromatic unit-distance graphs embedded in spheres, $G_{372} \subset S^2(r_1)$ and $G_{972} \subset S^2(r_2)$, where $r_1 = \cos \frac{\pi}{10}$ and $r_2 = \cos \frac{3\pi}{10}$, proving $\chi(S^2(r_1)) \geq 5$ and $\chi(S^2(r_2)) \geq 5$. They remark that $\chi(G_{372}) = 5$ is preserved after deleting any single vertex and that they “do not give here the subgraphs of G_{372} that have the property of minimality” [10, Sect. 4]. Cherkashin and Voronov [2] ask, more generally, for the minimal number of vertices of a subgraph G of $S^2(r)$ with $\chi(G) = \chi(S^2(r))$. We reduce both constructions.

Theorem 1. (a) *The graph $H_{231} \subset G_{372}$ defined in Section 3 is a unit-distance graph in $S^2(r_1)$ with 231 vertices and 938 edges, and $\chi(H_{231}) = 5$.*

(b) *The graph $H_{961} \subset G_{972}$ defined in Section 3 is a unit-distance graph in $S^2(r_2)$ with 961 vertices and 4028 edges, and $\chi(H_{961}) = 5$.*

Both graphs are vertex-critical, i.e. deleting any vertex leaves a 4-colourable graph, and neither contains the Moser spindle as a subgraph.

Thus H_{231} has 38% fewer vertices than G_{372} , while G_{972} turns out to be nearly vertex-critical (Remark 3). We do not know whether 231 is the least order of a 5-chromatic subgraph of G_{372} ; it is the smallest our search found. Since G_{372} and G_{972} are defined exactly in [10], our graphs are specified exactly by listing which of their vertices are kept or removed (Section 3 and Appendix A).

2 The parent graphs

Let Γ be the full symmetry group of the regular icosahedron (order 120, including reflections), generated as in [9] by

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \frac{1}{2} \begin{pmatrix} 1 & -\tau & \tau^{-1} \\ \tau & \tau^{-1} & -1 \\ \tau^{-1} & 1 & \tau \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \tau = \frac{1+\sqrt{5}}{2}.$$

G_{372} . Its vertices are the 12 vertices of the icosahedron with unit edge (the Γ -orbit of $(\frac{\tau}{2}, \frac{1}{2}, 0)$) together with the Γ -orbits of the three points $v_1, v_2, v_3 \in S^2(r_1)$ of Table 1, each orbit of size 120; edges join vertices at distance 1. It has 372 vertices and 1710 edges [10].

	value	minimal polynomial of twice the coordinate
x_1	0.71584...	$x^4 - 2x^3 + 2x^2 - x - 1$
y_1	0.34924...	$x^4 + x^3 - 2x^2 + 2x - 1$
z_1	0.51972...	$x^8 - 5x^6 + 2x^4 + 20x^2 - 19$
x_2, x_3	-0.07961..., -0.17646...	$x^8 - x^7 - 4x^6 + 6x^5 + 28x^4 - 7x^3 + 9x^2 + 8x + 1$
y_2, y_3	-0.08345..., 0.79929...	$x^8 - 2x^7 + x^6 - 5x^5 + 9x^4 - 4x^3 + 5x^2 - 5x - 1$
z_2	0.94404...	$x^8 - 3x^7 + 10x^5 + 4x^4 - 24x^3 - 13x^2 + 7x - 1$
z_3	0.48426...	$x^8 + 3x^7 - 10x^5 + 4x^4 + 24x^3 - 13x^2 - 7x - 1$

Table 1: Orbit representatives $v_i = (x_i, y_i, z_i)$ of G_{372} , from [10].

G_{972} . Its vertices are the 12 vertices of the great icosahedron with unit edge (the Γ' -orbit of $\tau^{-1}(\frac{1}{2}, \frac{\tau}{2}, 0)$, where Γ' is Γ conjugated by the coordinate swap $y \leftrightarrow z$, as in [9]) together with the Γ' -orbits of eight points $w_1, \dots, w_8 \in S^2(r_2)$, each of size 120. The coordinates of w_i are given in [9] as roots of explicit polynomials of degree 8 (reproduced in the repository, `construction/build.G972.py`);

for instance the second coordinate of w_8 is $\frac{1}{4}(1 - \sqrt{6 - 3\sqrt{5}} + 2\sqrt{(17\sqrt{5} - 37)/2})$. The graph has 972 vertices and 4110 edges.

Lemma 2. *For each sphere, all coordinates of all vertices of the parent graph lie in a number field K of degree 8 whose Galois closure has group D_8 (of order 16). Consequently K is a tower of quadratic extensions $\mathbb{Q} \subset \mathbb{Q}(\sqrt{5}) \subset \mathbb{Q}(\sqrt{5}, b) \subset \mathbb{Q}(\sqrt{5}, b, g) = K$ with $b = \sqrt{\beta}$, $g = \sqrt{\gamma}$, where*

$$\begin{aligned} S^2(r_1): \quad \beta &= \frac{7}{2} + \frac{5}{2}\sqrt{5}, & \gamma &= \frac{15}{4} - \frac{7}{4}\sqrt{5} + \left(\frac{3}{4} - \frac{1}{4}\sqrt{5}\right)b, \\ S^2(r_2): \quad \beta &= \frac{23}{2} + \frac{11}{2}\sqrt{5}, & \gamma &= \frac{5}{4} - \frac{3}{4}\sqrt{5} + \left(-\frac{1}{4} + \frac{1}{4}\sqrt{5}\right)b, \end{aligned}$$

all square roots being positive. Every coordinate of every vertex is a $\mathbb{Q}$ -linear combination of $1, \sqrt{5}, b, \sqrt{5}b, g, \sqrt{5}g, bg, \sqrt{5}bg$, i.e. an explicit nested radical with rational coefficients. For example, vertex 13 of H_{231} (Appendix A) is

$$\left(\frac{1}{4} + \frac{\sqrt{5}-1}{8}b, \quad -\frac{1+\sqrt{5}}{8} + \frac{1}{4}b, \quad \frac{1+\sqrt{5}}{4}g\right) \quad \text{on } S^2(r_1).$$

The field and the tower were computed with PARI/GP [6]; the exact coordinates of all vertices of H_{231} and H_{961} in this form, written with integers, fractions and square roots only, are in the repository (Mathematica and Python syntax).

3 The graphs H_{231} and H_{961}

The exact coordinates of every vertex are given as nested radicals (Lemma 2) in the repository. For readability, the tables below list coordinates rounded to four decimals; since the vertices of G_{372} are pairwise at distance at least 0.049, and those of G_{972} at least 0.009, these rounded triples identify the exact vertices unambiguously. Exact radical forms of all orbit representatives are given in Appendix B, so every vertex is defined exactly within this paper.

H_{231} is the subgraph of G_{372} induced by the 231 vertices listed in Appendix A: all 12 icosahedron vertices, 99 vertices of the orbit of v_1 , and 60 vertices of each of the orbits of v_2 and v_3 . Its symmetry group within Γ is trivial.

H_{961} is obtained from G_{972} by deleting the 11 vertices of Table 2.

	orbit	x	y	z
1	w_3 -orbit	-0.1495	-0.4014	0.4025
2	w_4 -orbit	0.5257	0.1359	-0.2250
3	w_4 -orbit	-0.1650	0.5628	-0.0388
4	w_4 -orbit	0.1228	0.3849	0.4269
5	w_4 -orbit	-0.1599	0.2878	0.4869
6	w_4 -orbit	0.2250	-0.5257	-0.1359
7	w_4 -orbit	0.4869	-0.1599	0.2878
8	w_4 -orbit	0.0600	0.4237	-0.4029
9	w_4 -orbit	-0.3849	-0.4269	-0.1228
10	w_4 -orbit	0.5628	0.0388	0.1650
11	w_4 -orbit	-0.4237	-0.4029	-0.0600

Table 2: The 11 vertices of G_{972} deleted to obtain H_{961} (four-decimal labels).

Table 3 summarizes both graphs and Figure 1 shows H_{231} .

graph	sphere	$ V $	$ E $	degrees	composition by orbit	vertex-critical
G_{372} [10]	$S^2(r_1)$	372	1710		$12 + 120 + 120 + 120$	no
H_{231}	$S^2(r_1)$	231	938	4–25 (avg. 8.12)	$12 + 99 + 60 + 60$	yes
G_{972} [10]	$S^2(r_2)$	972	4110		$12 + 8 \times 120$	no
H_{961}	$S^2(r_2)$	961	4028	5–65 (avg. 8.38)	$12 + 8 \times 120 - 11$	yes

Table 3: The parent graphs and the reduced graphs.

4 Proof of Theorem 1

The proof has four parts; parts (ii)–(iv) are finite computations, each accompanied by a certificate that can be checked independently of the software that produced it.

(i) Embedding and edges. By Lemma 2 all vertices have coordinates in a field K of degree 8. In PARI/GP [6] we represent K as $\mathbb{Q}[y]/(F(y))$, compute each coordinate of every orbit representative as an element of K (the roots of the minimal polynomials in K , choosing the real embedding of

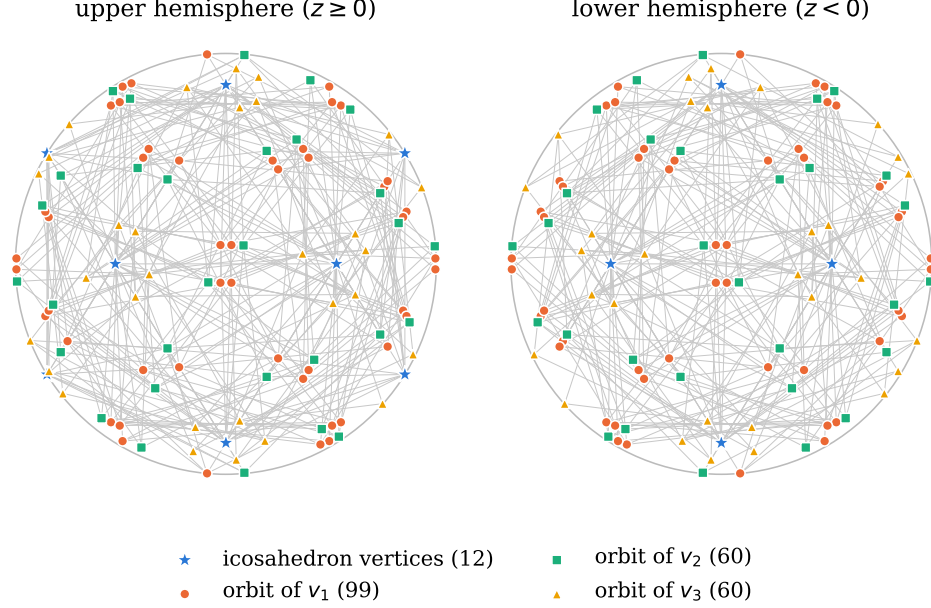


Figure 1: Orthogonal projections of H_{231} onto the xy -plane: vertices with $z \geq 0$ (left) and $z < 0$ (right, viewed from below). Marker shape and colour indicate the orbit of G_{372} a vertex belongs to; edges between vertices of the same hemisphere are drawn in grey. The bounding circle has radius r_1 .

K that matches the numerical values), generate the 120 matrices of Γ (resp. Γ') exactly, and form all vertices. For every vertex p of H_{231} and H_{961} we check $p \cdot p = r^2$ exactly, with $r_1^2 = \frac{5+\sqrt{5}}{8}$ and $r_2^2 = \frac{5-\sqrt{5}}{8}$, and for every edge pq we check $|p - q|^2 = 1$ exactly (938 and 4028 identities in K ; all hold). Conversely, no unit distance is missing: with the coordinates to 30 significant digits (files `H231_vertices.txt`, `H961_vertices.txt`), every non-adjacent pair satisfies $||p - q| - 1| \geq 2.2 \cdot 10^{-3}$ in H_{231} and $\geq 4.1 \cdot 10^{-4}$ in H_{961} , far above the numerical error. Hence both graphs are unit-distance graphs on the stated spheres with the stated numbers of edges.

This was double-checked independently of PARI/GP: a short program implements exact arithmetic in the basis $1, \sqrt{5}, b, \sqrt{5}b, g, \sqrt{5}g, bg, \sqrt{5}bg$ with rational coefficients (using only the relations $\sqrt{5}^2 = 5, b^2 = \beta, g^2 = \gamma$), reads the nested-radical coordinates from the files `H231_exact_coordinates.m` and `H961_exact_coordinates.m`, and confirms $p \cdot p = r^2$ for all vertices and $|p - q|^2 = 1$ for all 938 and 4028 edges; as a negative control, perturbing a single coordinate by 10^{-3} is detected (the vertex leaves the sphere and all its edges fail). The edges also form a subset of the edges of G_{372} and G_{972} , whose unit distances were verified symbolically in [10, 9].

(ii) No proper 4-colouring. We encode the existence of a proper 4-colouring as a CNF formula with variables $x_{v,c}$, clauses $\bigvee_c x_{v,c}$ and $\neg x_{v,c} \vee \neg x_{v,c'}$ ($c \neq c'$) for every vertex v , clauses $\neg x_{u,c} \vee \neg x_{v,c}$ for every edge uv and colour c , and two unit clauses fixing the colours of the end-points of one edge (which loses no generality). The SAT solver KISSAT [1] reports both formulas unsatisfiable and emits proofs in the DRAT format; the independent checker DRAT-TRIM [11] validates both proofs (s VERIFIED; files `H231_4col.cnf`, `H231_4col.drat.xz` and likewise for H_{961}). As further independent confirmation, the solvers CaDiCaL, Glucose and MiniSat (via PySAT [5]) and a second encoding with two Boolean variables per vertex give the same answer

(`verification/crosscheck_solvers.py`), and for H_{231} a plain backtracking search (DSATUR vertex order, colour-symmetry breaking, about $7 \cdot 10^6$ nodes, no SAT solver) finds no proper 4-colouring. Hence $\chi \geq 5$.

(iii) A proper 5-colouring. A 5-colouring of H_{231} is listed in Appendix A (colour classes of sizes 55, 51, 49, 46, 30); both colourings are in the files `H231_5colouring.txt` and `H961_5colouring.txt` and were checked edge by edge. Hence $\chi = 5$.

(iv) Criticality and spindles. For each vertex v the files `H231_criticality_certificates.txt` and `H961_criticality_certificates.txt` contain a proper 4-colouring of the graph with v deleted (231 and 961 certificates); each was checked edge by edge. Hence both graphs are vertex-critical. An exhaustive search over all pairs of rhombi (pairs of triangles sharing an edge) with a common apex whose tips are adjacent finds no Moser spindle in either graph. $\square$

5 How the graphs were found

Encoding with selectors. To test many vertex subsets $U \subseteq V$ we add a selector literal s_v to every at-least-one-colour clause, $\neg s_v \vee \bigvee_c x_{v,c}$, and solve under the assumptions $\{s_v : v \in U\}$, with the colours of one triangle of U fixed. An unsatisfiable answer comes with a *core*: a subset of the assumptions that is already unsatisfiable, i.e. a non-4-colourable subset of U .

Algorithm 1 (random-order vertex deletion).

1. Set $U \leftarrow V$ and $C \leftarrow \emptyset$ (vertices known to be essential).
2. Repeatedly remove vertices of degree < 4 in $G[U]$ (they never affect 4-colourability).
3. Pick a random $v \in U \setminus C$. If $G[U \setminus \{v\}]$ is not 4-colourable, replace U by the returned core (and go to step 2); otherwise add v to C .
4. Stop when $C = U$.

Each test was run with a conflict limit and an inconclusive test was treated as “4-colourable”, so the output is vertex-critical only up to these limits; criticality of the final graphs H_{231} and H_{961} was verified separately (Section 4). For G_{372} each test takes well under a second. Sixteen runs with different random orders gave non-4-colourable subgraphs with between 235 and 337 vertices (Figure 2).

Algorithm 2 (local search). Starting from a vertex-critical U : add to U between 5 and 40 random vertices of $V \setminus U$ having at least 2 to 4 neighbours in U , optionally delete a few random vertices of U , run Algorithm 1 on the resulting set, and accept the result if it is not larger than U . Starting from the 235-vertex subgraph this produced the 231-vertex graph H_{231} .

Triage for G_{972} . For G_{972} each test takes several seconds, so we first tested every vertex v of a 5-chromatic subgraph U with $|U| = 968$ separately. If $G[U \setminus \{v\}]$ is 4-colourable, then v lies in every non-4-colourable subgraph of $G[U]$ and need never be tested again. Only 34 of the 968 vertices were individually removable (931 were shown to be essential; the three vertices of the triangle whose colours were fixed were not tested); deleting the 34 greedily gave 963 vertices, and Algorithm 2 gave 961.

Remark 3. Unlike G_{372} , the graph G_{972} is close to vertex-critical: at least 931 of the 968 vertices of the first subgraph examined are essential.

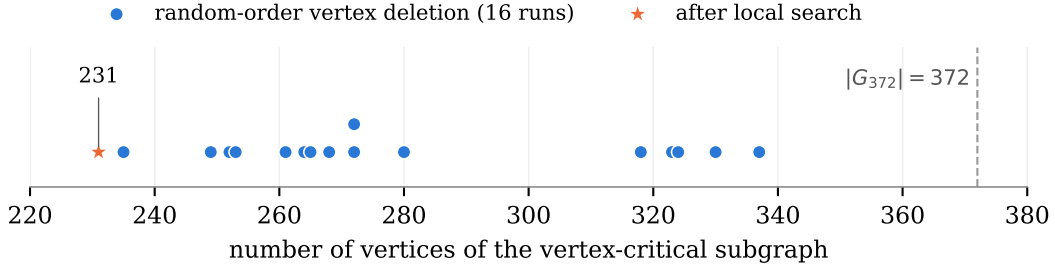


Figure 2: Sizes of the non-4-colourable subgraphs of G_{372} found by 16 runs of Algorithm 1 with different random orders, and the result of Algorithm 2.

6 Data availability

All data, proofs and code are in the repository accompanying this paper (<https://github.com/chandanshukla1989/5chromatic-spheres>): for both graphs the vertex lists (parent-graph indices and 30-digit coordinates), exact coordinates in nested radicals (Mathematica and Python syntax), edge lists, 5-colourings, criticality certificates, CNF formulas with compressed DRAT proofs and DRAT-TRIM logs, the scripts that rebuild G_{372} and G_{972} , and the verification programs. A single script (`verify_all.sh`) re-runs every check of Section 4. The Supplementary Material describes the files and procedures and lists all coordinates in radicals.

7 Concluding remarks

The graph H_{231} is a local optimum of our search; determining the smallest 5-chromatic subgraph of G_{372} , or the smallest 5-chromatic unit-distance graph in $S^2(r_1)$, remains open. An exact implicit-hitting-set computation of the former did not finish within the time we allotted. The other constructions of [10] may also contain much smaller 5-chromatic subgraphs.

Use of AI tools

The author initiated and led this work, set its verification standards (exact algebraic coordinates and certificate-based proofs) and takes full responsibility for all claims. AI tools were used for computational exploration, for implementing the search and verification scripts, and for drafting the text. Every claim rests on certificates or exact computations that can be re-checked independently (Section 4).

References

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A Vertex list and a 5-colouring of H_{231}

Each row gives an index, the orbit of G_{372} (I: icosahedron vertices; 1, 2, 3: orbits of v_1, v_2, v_3), the coordinates rounded to four decimals (labels of exact vertices of G_{372} , see Section 3), and a colour 1–5 of a proper 5-colouring.

#	orb.	x	y	z	col.
1	I	0.8090	0.5000	0.0000	1
2	I	−0.8090	−0.5000	0.0000	5
3	I	0.0000	0.8090	0.5000	4
4	I	0.8090	−0.5000	0.0000	3
5	I	0.0000	−0.8090	−0.5000	1
6	I	−0.8090	0.5000	0.0000	1
7	I	0.0000	−0.8090	0.5000	4
8	I	0.5000	0.0000	0.8090	2
9	I	−0.5000	−0.0000	0.8090	3
10	I	0.0000	0.8090	−0.5000	3
11	I	−0.5000	0.0000	−0.8090	4
12	I	0.5000	0.0000	−0.8090	2
13	1	0.7158	0.3492	0.5197	2
14	1	−0.7158	−0.3492	0.5197	5
15	1	0.5197	0.7158	0.3492	2
16	1	0.2360	0.4272	0.8163	2
17	1	0.5197	−0.7158	−0.3492	3
18	1	−0.5197	−0.7158	0.3492	3
19	1	0.3492	0.5197	0.7158	2
20	1	−0.2113	0.4671	0.8011	2
21	1	0.5197	−0.7158	0.3492	2
22	1	0.8163	0.2360	0.4272	3
23	1	0.0246	−0.0852	0.9469	1

#	orb.	x	y	z	col.
59	1	0.7311	-0.3739	-0.4799	5
60	1	-0.8011	0.2113	0.4671	2
61	1	-0.0852	-0.9469	0.0246	2
62	1	-0.0246	-0.0852	-0.9469	5
63	1	-0.9469	0.0246	0.0852	2
64	1	0.0246	-0.0852	-0.9469	2
65	1	-0.4272	0.8163	-0.2360	1
66	1	0.4671	0.8011	-0.2113	5
67	1	0.4799	0.7311	0.3739	4
68	1	0.3739	-0.4799	-0.7311	5
69	1	0.4671	-0.8011	-0.2113	5
70	1	-0.7158	0.3492	-0.5197	3
71	1	0.3739	0.4799	-0.7311	2
72	1	-0.2360	0.4272	-0.8163	2
73	1	0.8011	0.2113	-0.4671	1
74	1	0.7158	-0.3492	-0.5197	5
75	1	0.4671	-0.8011	0.2113	4
76	1	0.9469	-0.0246	0.0852	4
77	1	-0.0246	-0.0852	0.9469	1
78	1	0.7158	0.3492	-0.5197	1
79	1	0.7311	-0.3739	0.4799	3
80	1	0.2113	0.4671	0.8011	3
81	1	0.8011	-0.2113	-0.4671	5
82	1	-0.4671	0.8011	-0.2113	3
83	1	-0.9469	-0.0246	0.0852	2
84	1	-0.8163	0.2360	0.4272	5
85	1	-0.3739	0.4799	0.7311	1
86	1	0.4272	-0.8163	0.2360	3
87	1	0.7311	0.3739	0.4799	4
88	1	0.9469	-0.0246	-0.0852	1
89	1	-0.8163	-0.2360	-0.4272	5
90	1	0.2113	-0.4671	-0.8011	4
91	1	-0.3739	-0.4799	0.7311	5
92	1	-0.4671	-0.8011	0.2113	4
93	1	-0.4799	-0.7311	-0.3739	1
94	1	-0.9469	0.0246	-0.0852	2
95	1	-0.7158	-0.3492	-0.5197	3
96	1	-0.2113	-0.4671	-0.8011	1
97	1	-0.7311	0.3739	-0.4799	1
98	1	-0.4799	0.7311	-0.3739	1
99	1	-0.0246	0.0852	-0.9469	4
100	1	-0.4671	-0.8011	-0.2113	1
101	1	-0.3739	0.4799	-0.7311	3
102	1	-0.8011	-0.2113	-0.4671	2
103	1	-0.3739	-0.4799	-0.7311	1
104	1	-0.2360	-0.4272	-0.8163	3
105	1	-0.4671	0.8011	0.2113	1
106	1	0.0852	0.9469	-0.0246	3
107	1	0.4799	-0.7311	-0.3739	2
108	1	0.0852	-0.9469	-0.0246	2
109	1	0.4799	0.7311	-0.3739	1
110	1	0.2113	0.4671	-0.8011	2
111	1	-0.8011	0.2113	-0.4671	4
112	2	-0.0796	-0.0834	0.9440	5
113	2	0.0796	0.0834	0.9440	1
114	2	0.9440	0.0796	0.0834	1
115	2	-0.0834	0.9440	-0.0796	5
116	2	0.9440	0.0796	-0.0834	1

#	orb.	x	y	z	col.
117	2	0.8301	-0.2640	0.3818	3
118	2	0.3194	0.5622	0.6974	5
119	2	0.1844	-0.5106	0.7809	2
120	2	-0.9440	-0.0796	0.0834	2
121	2	0.0834	0.9440	0.0796	2
122	2	-0.2640	0.3818	0.8301	4
123	2	0.6974	-0.3194	0.5622	5
124	2	0.3990	-0.4334	0.7466	2
125	2	-0.5622	-0.6974	-0.3194	5
126	2	-0.9440	-0.0796	-0.0834	2
127	2	-0.8301	0.2640	0.3818	1
128	2	-0.0834	-0.9440	-0.0796	1
129	2	0.1844	0.5106	0.7809	3
130	2	0.5106	-0.7809	0.1844	4
131	2	0.3818	0.8301	0.2640	4
132	2	-0.3194	-0.5622	0.6974	1
133	2	-0.6974	-0.3194	-0.5622	4
134	2	0.3818	0.8301	-0.2640	4
135	2	0.6974	0.3194	0.5622	5
136	2	0.5622	0.6974	0.3194	5
137	2	0.7809	0.1844	-0.5106	1
138	2	-0.3818	-0.8301	-0.2640	3
139	2	0.0834	-0.9440	0.0796	3
140	2	0.4334	-0.7466	0.3990	3
141	2	-0.2640	-0.3818	0.8301	4
142	2	0.8301	-0.2640	-0.3818	5
143	2	0.5622	0.6974	-0.3194	3
144	2	0.7809	0.1844	0.5106	1
145	2	-0.3990	0.4334	0.7466	3
146	2	0.7466	0.3990	-0.4334	4
147	2	0.7466	-0.3990	-0.4334	4
148	2	-0.3194	-0.5622	-0.6974	1
149	2	-0.0796	-0.0834	-0.9440	4
150	2	0.4334	-0.7466	-0.3990	2
151	2	-0.5106	0.7809	0.1844	1
152	2	-0.3818	-0.8301	0.2640	4
153	2	-0.6974	0.3194	-0.5622	4
154	2	-0.7466	-0.3990	0.4334	3
155	2	-0.5622	-0.6974	0.3194	2
156	2	-0.7466	0.3990	0.4334	1
157	2	-0.7809	-0.1844	-0.5106	3
158	2	0.3990	-0.4334	-0.7466	2
159	2	0.0796	0.0834	-0.9440	4
160	2	0.5106	-0.7809	-0.1844	3
161	2	-0.4334	0.7466	0.3990	4
162	2	-0.8301	0.2640	-0.3818	1
163	2	-0.7809	-0.1844	0.5106	2
164	2	-0.1844	0.5106	-0.7809	3
165	2	-0.4334	0.7466	-0.3990	5
166	2	0.3194	0.5622	-0.6974	3
167	2	0.2640	0.3818	-0.8301	2
168	2	0.2640	-0.3818	-0.8301	5
169	2	-0.3990	0.4334	-0.7466	3
170	2	-0.5106	0.7809	-0.1844	1
171	2	-0.1844	-0.5106	-0.7809	1
172	3	-0.1765	0.7993	0.4843	1
173	3	0.1765	-0.7993	0.4843	4
174	3	0.4843	-0.1765	0.7993	4

#	orb.	x	y	z	col.
175	3	-0.4843	0.1765	0.7993	4
176	3	0.7993	0.4843	-0.1765	1
177	3	0.5852	0.1379	0.7369	2
178	3	0.0467	-0.8845	0.3464	4
179	3	0.4843	-0.1765	-0.7993	1
180	3	0.7081	-0.6319	-0.0624	3
181	3	-0.7993	0.4843	0.1765	3
182	3	0.7369	0.5852	0.1379	2
183	3	0.8845	0.3464	0.0467	1
184	3	-0.4088	0.1476	0.8459	3
185	3	-0.1379	-0.7369	0.5852	1
186	3	0.7993	-0.4843	-0.1765	3
187	3	0.6319	0.0624	0.7081	2
188	3	0.3464	0.0467	0.8845	3
189	3	0.5852	-0.1379	0.7369	2
190	3	0.7369	0.5852	-0.1379	1
191	3	0.3464	0.0467	-0.8845	3
192	3	0.8459	-0.4088	-0.1476	5
193	3	0.0467	0.8845	0.3464	4
194	3	-0.4843	0.1765	-0.7993	1
195	3	0.1379	0.7369	-0.5852	2
196	3	-0.7081	0.6319	-0.0624	1
197	3	-0.0624	0.7081	-0.6319	3
198	3	-0.3464	-0.0467	-0.8845	1
199	3	-0.7369	-0.5852	0.1379	2
200	3	-0.7993	-0.4843	0.1765	2
201	3	-0.4088	-0.1476	0.8459	1
202	3	-0.1476	-0.8459	-0.4088	2
203	3	-0.8845	-0.3464	0.0467	3
204	3	-0.0467	-0.8845	-0.3464	1
205	3	0.1379	0.7369	0.5852	2
206	3	0.8459	-0.4088	0.1476	1
207	3	0.7081	-0.6319	0.0624	3
208	3	-0.0624	-0.7081	-0.6319	2
209	3	-0.1765	0.7993	-0.4843	2
210	3	-0.1476	-0.8459	0.4088	1
211	3	-0.6319	-0.0624	0.7081	2
212	3	0.0624	0.7081	0.6319	4
213	3	-0.3464	-0.0467	0.8845	3
214	3	-0.8845	-0.3464	-0.0467	2
215	3	-0.7369	-0.5852	-0.1379	3
216	3	-0.8459	0.4088	-0.1476	4
217	3	-0.1379	-0.7369	-0.5852	4
218	3	0.1765	-0.7993	-0.4843	2
219	3	0.6319	0.0624	-0.7081	4
220	3	0.1476	0.8459	-0.4088	3
221	3	-0.0467	0.8845	-0.3464	2
222	3	0.0624	-0.7081	0.6319	4
223	3	-0.8459	0.4088	0.1476	1
224	3	-0.7081	0.6319	0.0624	4
225	3	0.1476	0.8459	0.4088	3
226	3	-0.5852	0.1379	-0.7369	1
227	3	0.8845	0.3464	-0.0467	4
228	3	0.4088	0.1476	-0.8459	4
229	3	0.4088	-0.1476	-0.8459	2
230	3	-0.6319	-0.0624	-0.7081	4
231	3	-0.5852	-0.1379	-0.7369	4

B Exact orbit representatives

With $b = \sqrt{\beta}$ and $g = \sqrt{\gamma}$ as in Lemma 2 (separately for each sphere, all square roots positive), the orbit representatives of G_{372} and G_{972} are as follows. Every vertex of H_{231} (resp. H_{961}) is Mv_i (resp. $M'w_i$) for a matrix $M \in \Gamma$ (resp. $M' \in \Gamma'$), or a vertex of the (great) icosahedron;

together with the generators of Γ in Section 2 this defines every vertex exactly within this paper. The four-decimal labels of Appendix A and Table 2 select the vertices.

v_1 (on $S^2(r_1)$)

$$\begin{aligned}x &= \frac{1}{4} - \frac{1}{8}b + \frac{1}{8}\sqrt{5}b \\y &= -\frac{1}{8} - \frac{1}{8}\sqrt{5} + \frac{1}{4}b \\z &= \frac{1}{4}g + \frac{1}{4}\sqrt{5}g\end{aligned}$$

v_2 (on $S^2(r_1)$)

$$\begin{aligned}x &= \frac{1}{16} + \frac{3}{16}\sqrt{5} - \frac{1}{16}b - \frac{1}{16}\sqrt{5}b + \frac{5}{16}g + \frac{1}{16}\sqrt{5}g - \frac{1}{8}bg \\y &= \frac{1}{8} - \frac{1}{16}b + \frac{1}{16}\sqrt{5}b + \frac{1}{16}g - \frac{1}{16}\sqrt{5}g - \frac{1}{16}bg - \frac{1}{16}\sqrt{5}bg \\z &= \frac{3}{16} + \frac{3}{16}\sqrt{5} - \frac{1}{8}b + \frac{1}{8}g + \frac{3}{16}bg + \frac{1}{16}\sqrt{5}bg\end{aligned}$$

v_3 (on $S^2(r_1)$)

$$\begin{aligned}x &= \frac{1}{16} + \frac{3}{16}\sqrt{5} - \frac{1}{16}b - \frac{1}{16}\sqrt{5}b - \frac{5}{16}g - \frac{1}{16}\sqrt{5}g + \frac{1}{8}bg \\y &= \frac{1}{8} - \frac{1}{16}b + \frac{1}{16}\sqrt{5}b - \frac{1}{16}g + \frac{1}{16}\sqrt{5}g + \frac{1}{16}bg + \frac{1}{16}\sqrt{5}bg \\z &= -\frac{3}{16} - \frac{3}{16}\sqrt{5} + \frac{1}{8}b + \frac{1}{8}g + \frac{3}{16}bg + \frac{1}{16}\sqrt{5}bg\end{aligned}$$

w_1 (on $S^2(r_2)$)

$$\begin{aligned}x &= -\frac{5}{8} + \frac{1}{8}\sqrt{5} + \frac{3}{8}b - \frac{1}{8}\sqrt{5}b - \frac{3}{16}g + \frac{1}{16}\sqrt{5}g - \frac{1}{16}bg + \frac{1}{16}\sqrt{5}bg \\y &= -\frac{1}{8} - \frac{1}{8}\sqrt{5} - \frac{1}{2}b + \frac{1}{4}\sqrt{5}b - \frac{1}{4}g + \frac{1}{8}\sqrt{5}g - \frac{3}{16}bg + \frac{1}{16}\sqrt{5}bg \\z &= -\frac{5}{8} + \frac{1}{8}\sqrt{5} + \frac{3}{16}g + \frac{1}{16}\sqrt{5}g + \frac{1}{4}bg - \frac{1}{8}\sqrt{5}bg\end{aligned}$$

w_2 (on $S^2(r_2)$)

$$\begin{aligned}x &= -\frac{5}{8} + \frac{1}{8}\sqrt{5} + \frac{3}{8}b - \frac{1}{8}\sqrt{5}b + \frac{3}{16}g - \frac{1}{16}\sqrt{5}g + \frac{1}{16}bg - \frac{1}{16}\sqrt{5}bg \\y &= -\frac{1}{8} - \frac{1}{8}\sqrt{5} - \frac{1}{2}b + \frac{1}{4}\sqrt{5}b + \frac{1}{4}g - \frac{1}{8}\sqrt{5}g + \frac{3}{16}bg - \frac{1}{16}\sqrt{5}bg \\z &= \frac{5}{8} - \frac{1}{8}\sqrt{5} + \frac{3}{16}g + \frac{1}{16}\sqrt{5}g + \frac{1}{4}bg - \frac{1}{8}\sqrt{5}bg\end{aligned}$$

w_3 (on $S^2(r_2)$)

$$\begin{aligned}x &= -\frac{159}{656} - \frac{69}{656}\sqrt{5} - \frac{23}{328}b + \frac{7}{164}\sqrt{5}b + \frac{3}{164}g - \frac{9}{328}\sqrt{5}g + \frac{29}{656}bg - \frac{23}{656}\sqrt{5}bg \\y &= -\frac{257}{656} + \frac{119}{656}\sqrt{5} + \frac{93}{656}b - \frac{37}{656}\sqrt{5}b - \frac{51}{656}g + \frac{15}{656}\sqrt{5}g - \frac{9}{82}bg + \frac{13}{328}\sqrt{5}bg \\z &= -\frac{87}{328} + \frac{7}{82}\sqrt{5} - \frac{113}{656}b + \frac{67}{656}\sqrt{5}b + \frac{17}{656}g - \frac{87}{656}\sqrt{5}g - \frac{17}{656}bg + \frac{5}{656}\sqrt{5}bg\end{aligned}$$

w_4 (on $S^2(r_2)$)

$$\begin{aligned}x &= -\frac{159}{656} - \frac{69}{656}\sqrt{5} - \frac{23}{328}b + \frac{7}{164}\sqrt{5}b - \frac{3}{164}g + \frac{9}{328}\sqrt{5}g - \frac{29}{656}bg + \frac{23}{656}\sqrt{5}bg \\y &= \frac{257}{656} - \frac{119}{656}\sqrt{5} - \frac{93}{656}b + \frac{37}{656}\sqrt{5}b - \frac{51}{656}g + \frac{15}{656}\sqrt{5}g - \frac{9}{82}bg + \frac{13}{328}\sqrt{5}bg \\z &= \frac{87}{328} - \frac{7}{82}\sqrt{5} + \frac{113}{656}b - \frac{67}{656}\sqrt{5}b + \frac{17}{656}g - \frac{87}{656}\sqrt{5}g - \frac{17}{656}bg + \frac{5}{656}\sqrt{5}bg\end{aligned}$$

w_5 (on $S^2(r_2)$)

$$\begin{aligned} x &= -\frac{169}{656} + \frac{69}{656}\sqrt{5} + \frac{23}{328}b - \frac{7}{164}\sqrt{5}b - \frac{3}{164}g + \frac{9}{328}\sqrt{5}g - \frac{29}{656}bg + \frac{23}{656}\sqrt{5}bg \\ y &= \frac{93}{656} + \frac{45}{656}\sqrt{5} - \frac{93}{656}b + \frac{37}{656}\sqrt{5}b + \frac{51}{656}g - \frac{15}{656}\sqrt{5}g + \frac{9}{82}bg - \frac{13}{328}\sqrt{5}bg \\ z &= -\frac{9}{82} + \frac{13}{328}\sqrt{5} + \frac{31}{656}b + \frac{15}{656}\sqrt{5}b + \frac{147}{656}g - \frac{77}{656}\sqrt{5}g + \frac{17}{656}bg - \frac{5}{656}\sqrt{5}bg \end{aligned}$$

w_6 (on $S^2(r_2)$)

$$\begin{aligned} x &= -\frac{169}{656} + \frac{69}{656}\sqrt{5} + \frac{23}{328}b - \frac{7}{164}\sqrt{5}b + \frac{3}{164}g - \frac{9}{328}\sqrt{5}g + \frac{29}{656}bg - \frac{23}{656}\sqrt{5}bg \\ y &= \frac{93}{656} + \frac{45}{656}\sqrt{5} - \frac{93}{656}b + \frac{37}{656}\sqrt{5}b - \frac{51}{656}g + \frac{15}{656}\sqrt{5}g - \frac{9}{82}bg + \frac{13}{328}\sqrt{5}bg \\ z &= \frac{9}{82} - \frac{13}{328}\sqrt{5} - \frac{31}{656}b - \frac{15}{656}\sqrt{5}b + \frac{147}{656}g - \frac{77}{656}\sqrt{5}g + \frac{17}{656}bg - \frac{5}{656}\sqrt{5}bg \end{aligned}$$

w_7 (on $S^2(r_2)$)

$$\begin{aligned} x &= -\frac{3}{8} + \frac{3}{16}b - \frac{1}{16}\sqrt{5}b - \frac{3}{16}g + \frac{1}{16}\sqrt{5}g + \frac{1}{16}bg - \frac{1}{16}\sqrt{5}bg \\ y &= \frac{1}{16} - \frac{1}{16}\sqrt{5} + \frac{1}{4}b - \frac{1}{8}\sqrt{5}b - \frac{1}{4}g + \frac{1}{8}\sqrt{5}g + \frac{3}{16}bg - \frac{1}{16}\sqrt{5}bg \\ z &= -\frac{7}{16} + \frac{1}{16}\sqrt{5} - \frac{1}{16}b + \frac{1}{16}\sqrt{5}b - \frac{5}{16}g + \frac{1}{16}\sqrt{5}g - \frac{1}{4}bg + \frac{1}{8}\sqrt{5}bg \end{aligned}$$

w_8 (on $S^2(r_2)$)

$$\begin{aligned} x &= -\frac{3}{8} + \frac{1}{8}\sqrt{5} + \frac{3}{16}g - \frac{1}{16}\sqrt{5}g + \frac{1}{16}bg - \frac{1}{16}\sqrt{5}bg \\ y &= \frac{1}{4} - \frac{1}{4}g + \frac{1}{8}\sqrt{5}g - \frac{3}{16}bg + \frac{1}{16}\sqrt{5}bg \\ z &= -\frac{1}{8} - \frac{1}{8}\sqrt{5} + \frac{3}{8}b - \frac{1}{8}\sqrt{5}b + \frac{5}{16}g - \frac{1}{16}\sqrt{5}g - \frac{1}{4}bg + \frac{1}{8}\sqrt{5}bg \end{aligned}$$